HW #2

1) a) 4.2 m/s b) 4.0 m/s c) −6.0 m/s

2) Let the direction batter is hitting towards, positive; the direction pitcher is throwing towards, negative.

$$a = \frac{40.0\ m/s - (-)31.0\ m/s}{4.00 \times 10^{-2}\ s}$$

$$= 1775\ m/s^2 \quad = 1.775 \times 10^3\ m/s^2$$

3)

a) $V(t) = \frac{d}{dt}\ 3.00t^2 - 5.00t + 4.00$

$= 6.00t - 5.00 \quad \rightarrow t = 3.00\ s \quad = 18.00 - 5.00$

$a(t) = \frac{d}{dt}\ 6.00t - 5.00 \quad = 13.00\ s$

$= 6.00 \quad \rightarrow t = 3.00\ s \quad = 6.00\ s$

b) $0 = 6.00t - 5.00$

$t = \frac{5.00}{6.00}\ s \quad \approx .83\ s$

4) a) DNE b) [3, 4] c) 1.5, 3.5̄ d) ~~[4, 7]~~ (4, 5.5), [5.5, 7]

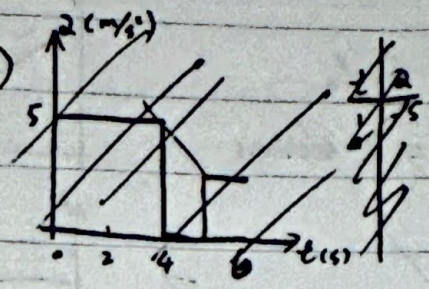
e) DNE. f) DNE g) [0, 1.5), (1.5, 3], [7, 8]

5) a) $[(2 \cdot -4 \cdot \frac{1}{2}) + (1 \cdot -4 \cdot \frac{1}{2}) + (1 \cdot 4 \cdot \frac{1}{2}) + (2 \cdot 4)]/6 = .67\ m/s$

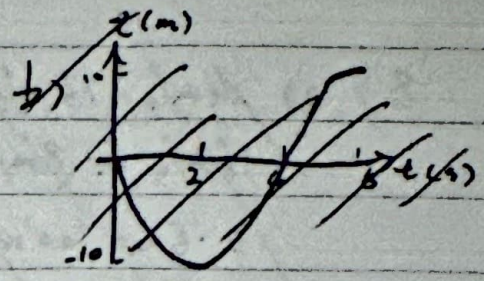
b) $[(2 \cdot 4 \cdot \frac{1}{2}) + (1 \cdot 4 \cdot \frac{1}{2}) + (1 \cdot 4 \cdot \frac{1}{2}) + (2 \cdot 4)]/6 = 2.67\ m/s$

6)

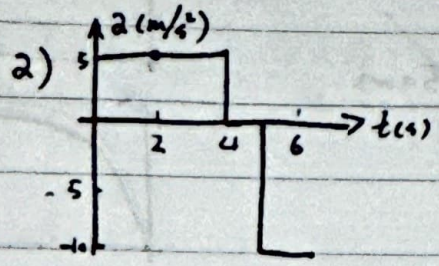
a)



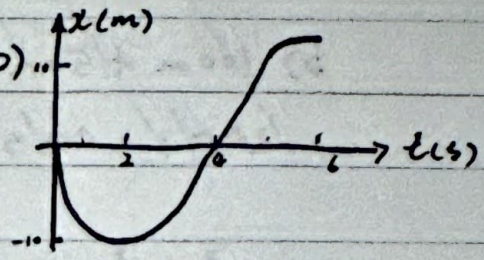
b)



a)



b)

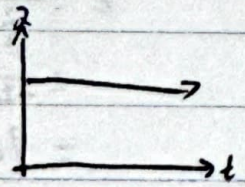


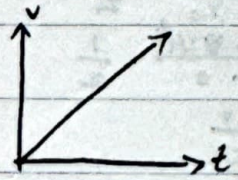
c) $[(5\cdot 4)+(0\cdot 1)+(-10\cdot 1)]/6$

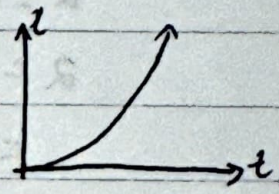
$= 1.67\ m/s^2$

d) $5\ m/s^2$

7) a)







$a = .500\ m/s^2$ $\quad v = .500t\ m/s$ $\quad x = .250t^2\ m$

a) $100 + 44.5m = .250t^2\ m$

$144.5m = \frac{1}{4}t^2\ m$

$144.5m \cdot 4 = t^2\ m$

$578.0 = t^2$

$\sqrt{578} = t$

$t = 24.042\ (s)$

$= 2.4042 \times 10\ s$

b) $100\ m = .250t^2\ m$

$100 \cdot 4 = t^2$

$\sqrt{400} = t$

$t = 20\ s$

$V_f = .500 \cdot 20\ m/s$

$= 10.000\ m/s$

$= 1.00 \times 10\ m/s$

$V_f = .500 \cdot 24.042\ m/s$

$= 12.021\ m/s$

$= 1.202 \times 10\ m/s$

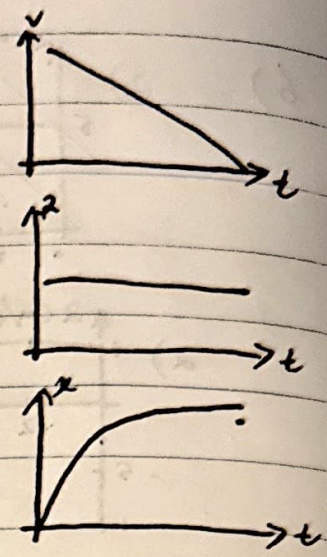
8) a) ~~24 m/s - 16 m/s = 8 m/s~~

~~16 m/s ÷ 8 m/s = 2 sections~~

~~2 · 50 m = 100 m~~

b) ~~100 m + 50 m = 150 m~~

~~24 m/s - 0 m/s =~~



a) $V^2 = V_0^2 + 2a(x - x_0)$

$$V^2 - V_0^2 = 2a(x - x_0)$$

$$2a = \frac{V^2 - V_0^2}{x - x_0}$$

$$a = \frac{V^2 - V_0^2}{x - x_0} \cdot \frac{1}{2}$$

$$a = \frac{16^2 - 24^2}{50 - 0} \cdot \frac{1}{2}$$

$$= \frac{-320}{100}$$

$$= -3.2\ m/s^2$$

$$V^2 - V_0^2 = 2a(x - x_0)$$

$$\frac{V^2 - V_0^2}{2a} = x - x_0$$

$$x = \frac{V^2 - V_0^2}{2a} + x_0$$

$$= \frac{0^2 - 16^2}{-2 \cdot 3.2} + 50$$

$$= 90\ m$$

b) ~~$x = x_1 + x_2$~~

~~$x = 90 + 50$~~

~~$x = 140$~~

$$V = V_0 + at$$

$$V - V_0 = at$$

$$t = \frac{V - V_0}{a}$$

$$= 7.5\ s$$